Year 12 Further Maths - Binomial and geometric distributions

Similar questions - Parameters and probabilities

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 **In this question you must show detailed reasoning.** The random variable T has a binomial distribution. It is known that $\mathrm{E}(T) = 7.2$ and the standard deviation of T is 1.2. Find the values of the parameters of the distribution. **[5]**

2 Two random variables X and Y have the distributions $\mathrm{B}(m, p)$ and $\mathrm{B}(n, p)$ respectively, where $p > 0$. It is known that

- $\mathrm{E}(Y) = 3\,\mathrm{E}(X)$,
- $\mathrm{Var}(Y) = 1.5\,\mathrm{E}(X)$.

(a) Determine the value of p. **[4]**

(b) Given also that $m = 4$, find $\mathrm{P}(Y = 6)$. **[3]**

3 A bookseller estimates that the probability that a delivery of books arrives on any one weekday is 0.15. The first weekday in October on which a delivery arrives is the Xth weekday of October.

(a) State an assumption needed for X to be well modelled by a geometric distribution. **[1]**

(b) Find $\mathrm{P}(X = 9)$. **[2]**

(c) Find $\mathrm{P}(X \leqslant 6)$. **[2]**

(d) Find $\mathrm{Var}(X)$. **[2]**

(e) Give a reason why a geometric distribution might not be appropriate for the first weekday of a calendar year on which a delivery arrives. **[1]**

4 The discrete random variable X has a geometric distribution. It is given that $\mathrm{Var}(X) = 30$. Determine $\mathrm{P}(X \geqslant 5)$. **[6]**

5 A bag contains a mixture of red and yellow counters in unknown proportions. The proportion of yellow counters in the bag is denoted by p.

(a) Sasha selects 12 counters at random with replacement. Write down an expression, in terms of p, for the variance of the number of yellow counters she selects. **[1]**

Fred selects one counter at random, notes its colour and replaces it. He continues until a yellow counter is selected. The first yellow counter is the Xth that Fred selects.

(b) Assume that $p = 0.4$. Find (i) $\mathrm{P}(X \geqslant 4)$, (ii) $\mathrm{Var}(X)$. **[3]**

(c) In fact, on the basis of a large number of observations of X, it is found that $\mathrm{P}(X = 4) = \frac{8}{27}\,\mathrm{P}(X = 1)$. Estimate the value of p. **[5]**

Question 1

$np = 7.2$ and $np(1-p) = 1.2^2 = 1.44$. Dividing: $1-p = \dfrac{1.44}{7.2} = 0.2$, so $p = 0.8$ and $n = \dfrac{7.2}{0.8} = 9$. So $T \sim \text{B}(9, 0.8)$

Question 2

(a) $np = 3mp$ so $n = 3m$. $np(1-p) = 1.5mp$, so $3mp(1-p) = 1.5mp$ and $1-p = 0.5$, $p = 0.5$

(b) $n = 12$: $\text{P}(Y = 6) = {}^{12}C_6(0.5)^{12} = \dfrac{924}{4096} = 0.226$

Question 3

(a) Deliveries on different days are independent (and the probability is constant)

(b) $\text{P}(X = 9) = 0.85^8(0.15) = 0.0409$

(c) $1 - 0.85^6 = 0.623$

(d) $\dfrac{0.85}{0.15^2} = 37.8$

(e) The probability of a delivery arriving may differ (e.g. after the holiday period, or depend on the day of the week), so the probability would not be constant

Question 4

$\dfrac{1-p}{p^2} = 30$, so $30p^2 + p - 1 = 0$ and $(6p-1)(5p+1) = 0$. $p = \frac{1}{6}$. $\text{P}(X \geqslant 5) = \left(\frac{5}{6}\right)^4 = 0.482$

Question 5

(a) $12p(1-p)$

(b)(i) $\text{P}(X \geqslant 4) = 0.6^3 = 0.216$. (ii) $\text{Var}(X) = \dfrac{0.6}{0.4^2} = 3.75$

(c) $(1-p)^3 p = \frac{8}{27}p$, so $(1-p)^3 = \frac{8}{27}$ and $1-p = \frac{2}{3}$, $p \approx \frac{1}{3}$